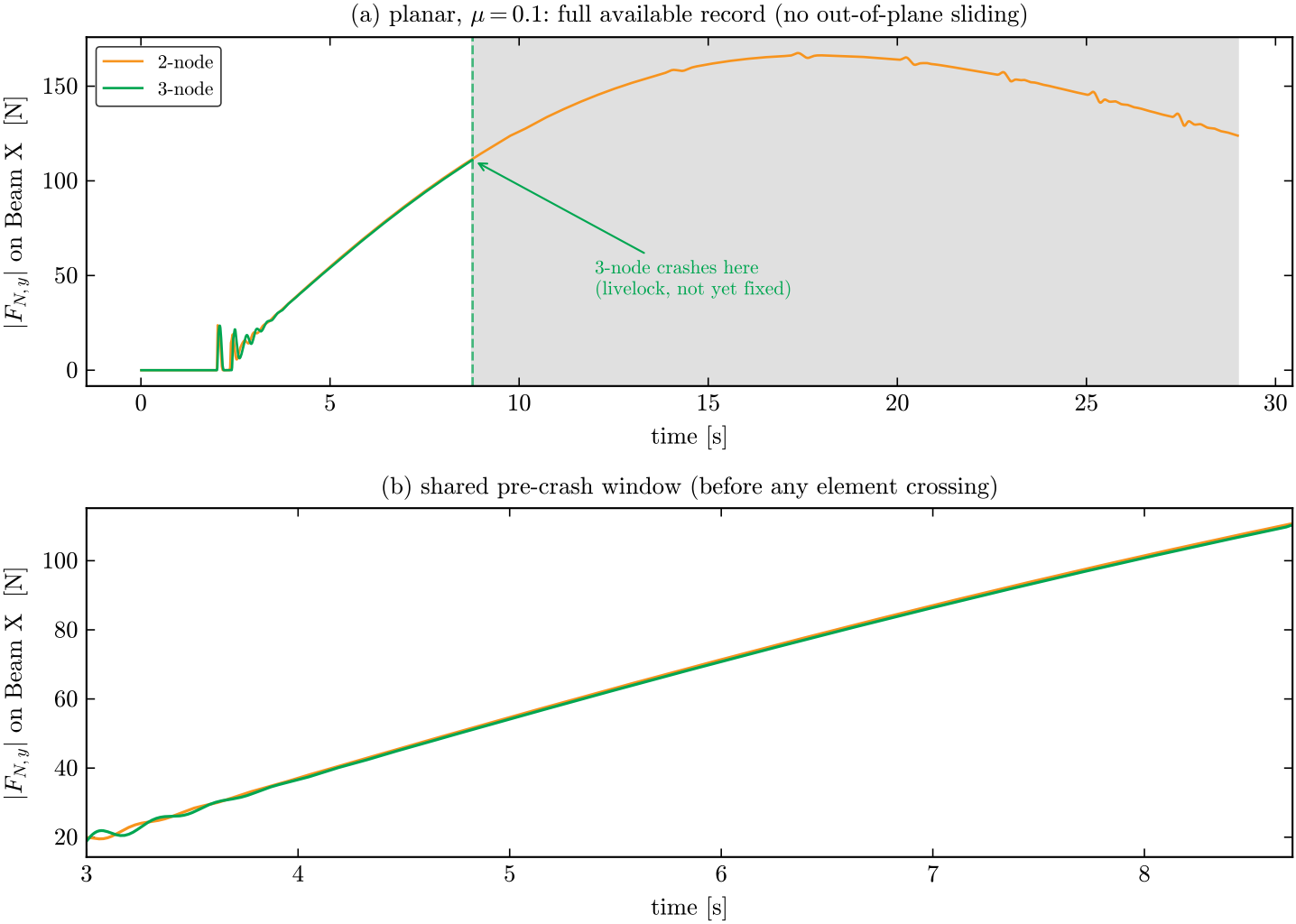
(a) planar, $\mu = 0.1$: full available record (no out-of-plane sliding)
2-node
3-node
3-node crashes here
(livelock, not yet fixed)
$|F_{N,y}|$ on Beam X [N]
time [s]
(b) shared pre-crash window (before any element crossing)
$|F_{N,y}|$ on Beam X [N]
time [s]